

Skills Summary

The Derivative as a Limit and as a Rate

Use this reference while completing your worksheet or when studying.

Skill 1 — The Derivative as a Limit of Slopes

Definition:

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

The fraction is the slope of the line through $(a, f(a))$ and $(a+h, f(a+h))$. Letting $h \rightarrow 0$ slides the second point into the first, and the secant slope becomes the tangent slope. You may never substitute $h = 0$ straight away — that is $\frac{0}{0}$. Simplify until the h downstairs cancels.

Example: $f(x) = x^2$. Find $f'(5)$.

Step 1. Substituting $h = 0$ in the difference quotient gives $\frac{0}{0}$, so expand the top first and cancel the h — that is the only move available.

Step 2. $\frac{(5+h)^2 - 25}{h} = \frac{25 + 10h + h^2 - 25}{h} = \frac{h(10+h)}{h} = 10+h$, and as $h \rightarrow 0$ that tends to 10.

$$\Rightarrow f'(5) = 10$$

Try it: $f(x) = x^2$. Find $f'(4)$ from the definition.

check: 8

Skill 2 — Average Rate Versus Instant Rate

Two different questions. Average rate over $[a, b]$ is $\frac{f(b)-f(a)}{b-a}$: one slope, between two points, describing a whole interval. Instantaneous rate at a is $f'(a)$: the slope of the curve at a single point. Read the wording — “from ... to ...” asks for the first, “at ...” asks for the second, and they are almost never equal.

Example: $s(t) = t^2$ metres at t seconds. Find the average velocity from $t = 2$ to $t = 5$.

Step 1. “From 2 to 5” names an interval, so this is the two-point slope, not a derivative — no limit is needed.

Step 2. $\frac{s(5) - s(2)}{5 - 2} = \frac{25 - 4}{3} = \frac{21}{3} = 7$ metres per second.

$$\Rightarrow 7 \text{ metres per second}$$

Try it: Same $s(t) = t^2$. Find the average velocity from $t = 1$ to $t = 4$.

check: 5 metres per second

Skill 3 — What a Derivative Value Means

Reading $f'(a)$ in three parts. The **sign** says the direction: positive means the output is increasing at that moment, negative means it is decreasing. The **size** says how fast. The **units** are always output units per input unit — litres per minute, metres per second, dollars per item. A derivative is a rate at an instant, never a total.

Example: A balloon's volume is $V(t) = t^2 + 1$ litres at t minutes. Find $V'(3)$ and say what it means.

Step 1. The question asks for a rate at one instant, so differentiate and then substitute — the derivative function is a machine that turns an input into a slope.

Step 2. $V'(t) = 2t$, so $V'(3) = 2(3) = 6$: positive, size 6, units litres per minute — the balloon is filling at 6 litres per minute at that instant.

$$\Rightarrow V'(t) = 2t \quad \text{and} \quad V'(3) = 6$$

Try it: $f(x) = x^2 + 5x$. Find $f'(x)$, then $f'(2)$.

6 per **5** + **2x** :kæp